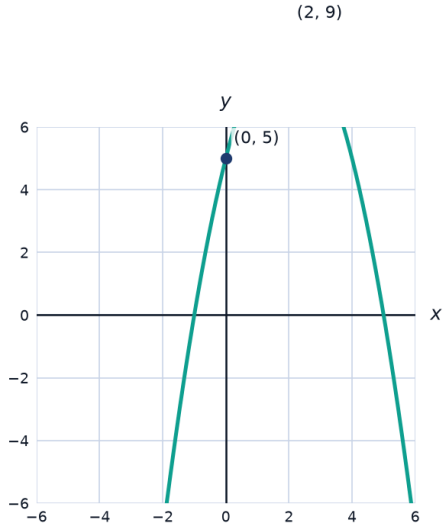
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

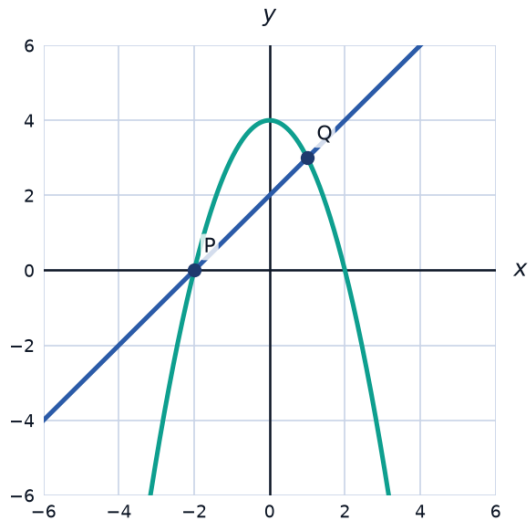
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

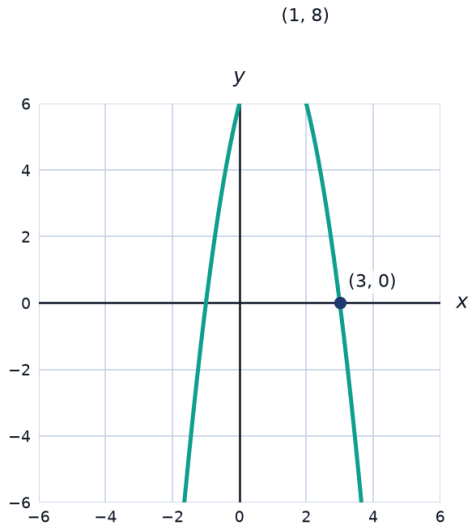
The graph of the polynomial function $f(x) = x^3 + ax^2 + bx - 6$ has a local maximum at $x = -3$ and a local minimum at $x = 1$. Find the values of a and b.



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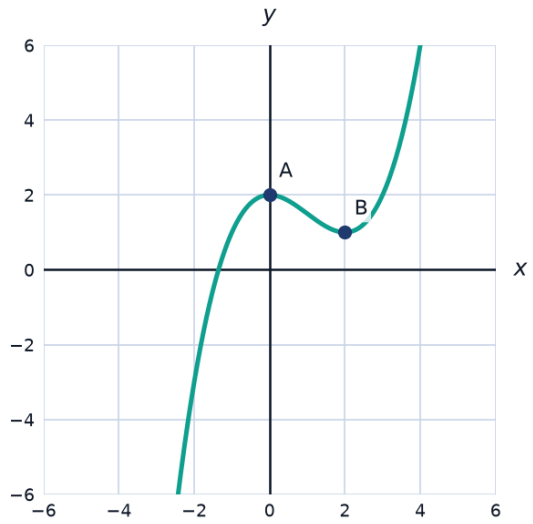
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?



4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?



5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(-1, 5)$. If the graph of $f(x)$ also passes through the point $(2, -13)$, find the value of b.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the polynomial function $f(x) = x^3 - ax^2 + bx - 6$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. Determine the value of $a + b$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ is a parabola that passes through the points $(-1, 5)$ and $(2, -7)$. If the vertex of the parabola is located at $x = -1/2$, find the value of c.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = ax^2 + bx - 5$. If the graph of $f(x)$ passes through the points $(-2, 3)$ and $(1, -3)$, find the value of $a - b$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the sinusoidal function $y = A\sin(Bx + C) + D$ is shown below. The function has a maximum value of 5 and a minimum value of -1. The graph passes through the point $(0, 2)$ and has a period of $\frac{\pi}{2}$. Find the value of A and B. Assume $A > 0$ and $B > 0$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $f(x) = a\sqrt{x - h} + k$ is shown below. The function passes through the points $(5, 10)$ and $(13, 16)$. Determine the value of a and the coordinates of the vertex (h, k).

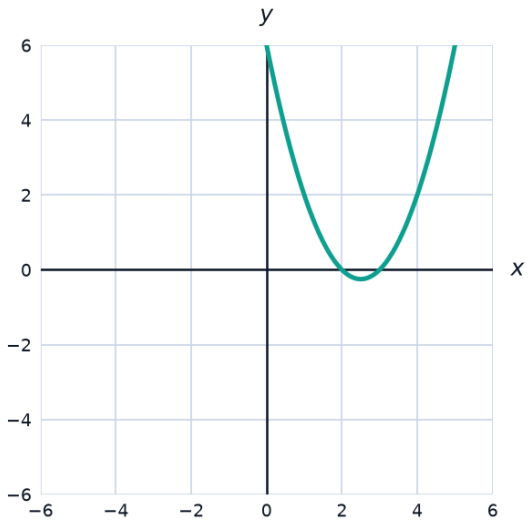
12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the rational function $f(x) = \frac{x-2}{x+3}$, what value of x is excluded from the domain?

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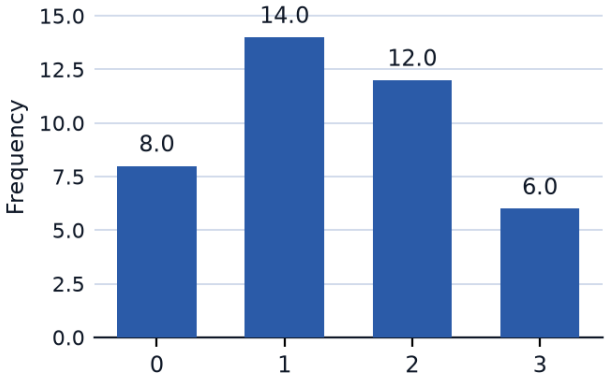
13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the rational function $f(x) = \frac{x^2 - 5x + 6}{x^2 - 7x + 12}$. Let $g(x)$ be the function obtained by simplifying $f(x)$. Determine the value of $g(x)$ as $x \to \infty$.



14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The bar graph shows the distribution of the number of goals scored per game by a soccer team across all 40 games of its regular season. If the team plays k post-season games and scores exactly 3 goals in each of those games, the mean number of goals scored per game across all regular-season and post-season games combined will be exactly 2. What is the value of k?

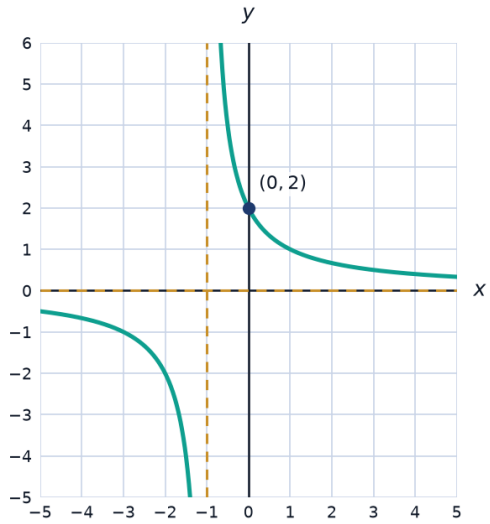


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Rational Expressions

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?



16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

What is the period of the trigonometric function $f(x) = 5\cos(4x) - 2$?

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of this function?

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

Determine all continuous even functions f satisfying $f(f(x)) = x$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Conic Sections

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Trigonometric Functions

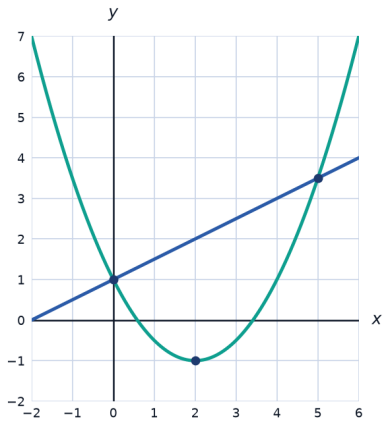
Consider the polynomial function $P(x) = 2x^3 - 5x^2 + 3x - 7$. What is the value of $P(2)$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in review: relations and functions, evaluate $f(2)$.



a. 7

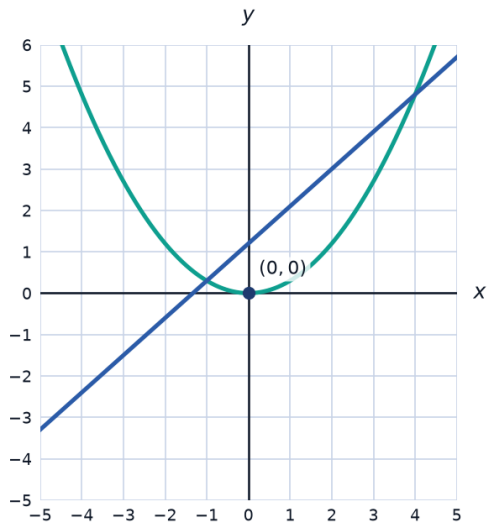
b. 9

c. 4

d. 11

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

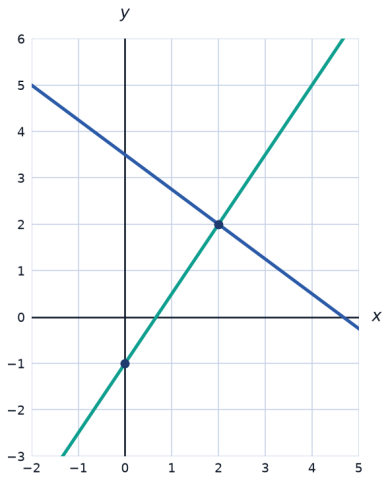
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in function families and transformations, evaluate $f(2)$.



a. 10

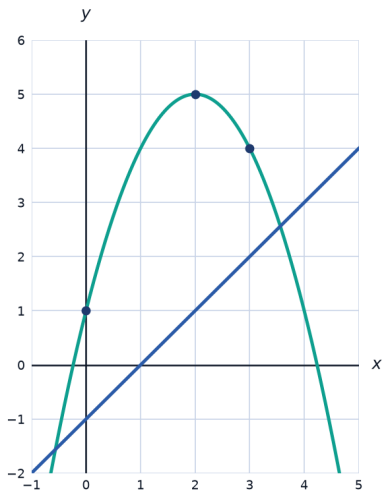
b. 13

c. 6

d. 14

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in choosing appropriate models, evaluate $f(2)$.



a. 13

b. 17

c. 8

d. 17

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: relations and functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for function families and transformations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for domain and range and continuity consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in domain and range and continuity, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for characteristics from a graph consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in characteristics from a graph, evaluate $f(4)$.

a. 31

b. 37

c. 24

d. 35

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 8$ in average rates of change, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

State and apply the intermediate value theorem: if $f(1) = -3$ and $f(4) = 5$, continuous on $[1, 4]$, then:

a. No conclusion possible

b. There exists $c \in (1, 4)$ with $f(c) = 0$

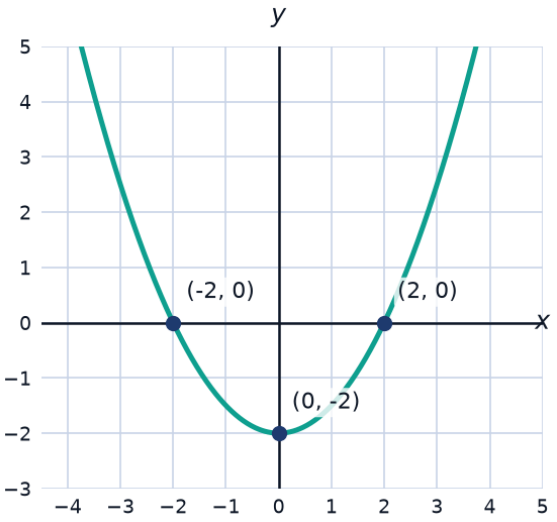
c. $f(2) = 1$

d. f is constant

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33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have?



a. 3

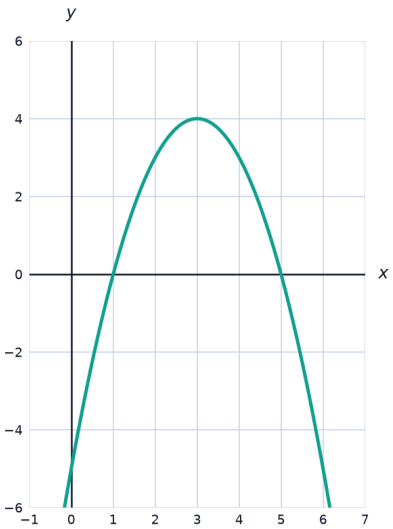
b. 4

c. 6

d. 8

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

The graph of the function $f(x) = -(x - 3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$?



a. -5

b. 3

c. 4

d. 7

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the sum of all real values of k for which the equation $|x^2 - 6x + 5| = 2x + k$ has exactly three distinct real solutions?

a. -3

b. -1

c. -2

d. -13

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = x^2 + 1$ and $g(x) = 3x - 2$. What is the value of $(f \circ g)(2)$?

a. 17

b. 13

c. 20

d. 16

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the number of real-valued functions $f : \{1, 2, 3\} \to \{1, 2, 3\}$ satisfying $f(f(x)) = f(x)$ for all x.

a. 9

b. 10

c. 27

d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Sequences & Series

Consider the function $f(x) = 2x + 5$. Which of the following represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{x-5}{2}$

b. $f^{-1}(x) = \frac{x+5}{2}$

c. $f^{-1}(x) = \frac{2}{x} - 5$

d. $f^{-1}(x) = 2x - 5$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Conic Sections

For $f : \mathbb{Z} \to \mathbb{Z}$, if $f(m + n) + f(mn - 1) = f(m)f(n) + 1$ for all integers, find $f(2)$.

a. 1

b. 3

c. 4

d. 2

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Trigonometric Functions

Consider the function $g(x) = x^2 - 6x + 5$. What is the x-coordinate of the vertex of the parabola represented by this function?

a. 3

b. -3

c. 6

d. -6

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Matrices & Systems

Solve: $f(x) = f(-x)$ for all x where $f(x) = x^3 + ax$. Find a.

a. $a = -1$

b. Any a

c. $a = 0$

d. $a = 1$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Probability & Statistics

Classify all solutions of $f(f(x)) = f(x)$ (idempotents) for $f(x) = x^2$.

a. $x = 1$ only

b. $x = 0$ or $x = 1$

c. All $x \geq 0$

d. $x = 0$ only

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Given $f(x) = \frac{1}{2 - x}$, find $f^{(n)}(x)$ (the n-th iterate).

a. $f^n(x) = nx + 1$

b. $f^n(x) = f(x)$ for all n

c. $f^n(x) \to \infty$

d. Cycles with period 3; $f^3(x) = x$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Polynomial Functions

Find all k such that $f(x) = kx^2 + (k+2)x + 2k - 1$ is always negative.

a. $k < -1$

b. All $k < 0$

c. $k < -\frac{1}{3}$ and discriminant < 0

d. No such k exists for all x

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Rational Expressions

Find the maximum value of $f(x) = x^a(1-x)^b$ on $[0, 1]$ where $a, b > 0$.

a. $\left(\frac{a}{a+b}\right)^a \left(\frac{b}{a+b}\right)^b$

b. $\frac{a^a b^b}{(a+b)^{a+b}}$

c. $\frac{1}{4}$ for $a = b = 1$

d. Both A and B

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